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Sulfur trioxide

Sulfur trioxide (alternative spelling **sulphur trioxide**, also known as *nisso sulfan*) is the chemical compound with the formula SO_3 . It has been described as "unquestionably the most important economically" sulfur oxide.^[1] It is prepared on an industrial scale as a precursor to sulfuric acid.

Sulfur trioxide exists in several forms - gaseous monomer, crystalline trimer, and solid polymer. Sulfur trioxide is a solid at just below room temperature with a relatively narrow liquid range. Gaseous SO_3 is the primary precursor to acid rain.^[6]

Molecular structure and bonding

Monomer

The molecule SO_3 is trigonal planar. As predicted by VSEPR theory, its structure belongs to the D_{3h} point group. The sulfur atom has an oxidation state of +6 and may be assigned a formal charge value as low as 0 (if all three sulfur-oxygen bonds are assumed to be double bonds) or as high as +2 (if the Octet Rule is assumed).^[7] When the formal charge is non-zero, the S-O bonding is assumed to be delocalized. In any case the three S-O bond lengths are equal to one another, at 1.42 Å.^[1] The electrical dipole moment of gaseous sulfur trioxide is zero.



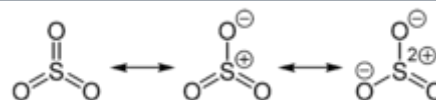
Ball-and-stick model of the cyclic trimer of SO_3

dependent on traces of water.^[9]

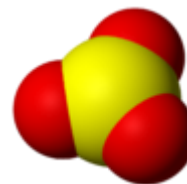
Trimer

Both liquid and gaseous^[8] SO_3 exists in an equilibrium between the monomer and the cyclic trimer. The nature of solid SO_3 is complex and at least 3 polymorphs are known, with conversion between them being

Sulfur trioxide



Resonance structures



Space-filling model^[1]

Names

Preferred IUPAC name

Sulfur trioxide

Systematic IUPAC name

Sulfonylideneoxidane

Other names

Sulfuric anhydride, Sulfur(VI) oxide

Identifiers

CAS Number

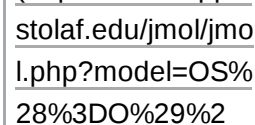
7446-11-9 (https://commonchemistry.cas.org/detail?cas_rn=7446-11-9) ✓

3D model (JSmol)

monomer:
[Interactive image \(https://chemapps.stolaf.edu/jmol/jmol.php?model=O%3DS%28%3DO%29%3DO\)](https://chemapps.stolaf.edu/jmol/jmol.php?model=O%3DS%28%3DO%29%3DO)
γ-trimer:
[Interactive image \(https://chemapps.stolaf.edu/jmol/jmol.php?model=O%3DSO%28%3DO%29OS%28%3DO%29\)](https://chemapps.stolaf.edu/jmol/jmol.php?model=O%3DSO%28%3DO%29OS%28%3DO%29)

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α/β polymer:
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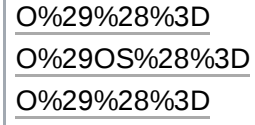
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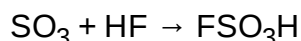
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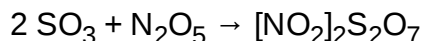
O%29%28%3D
O%29OS%28%3D
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Akin to the behavior of H_2O , hydrogen fluoride adds to give fluorosulfuric acid:



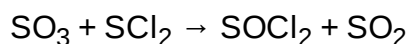
Deoxygenation

SO₃ reacts with dinitrogen pentoxide to give the nitronium salt of pyrosulfate:



Oxidant

Sulfur trioxide is an oxidant. It oxidizes sulfur dichloride to thionyl chloride.



Lewis acid

SO₃ is a strong Lewis acid readily forming adducts with Lewis bases.^[13] With pyridine, it give the sulfur trioxide pyridine complex. Related adducts form from dioxane and trimethylamine.

Sulfonating agent

Sulfur trioxide is a potent sulfonating agent, i.e. it adds SO₃ groups to substrates. Often the substrates are organic, as in aromatic sulfonation.^[14] For activated substrates, Lewis base adducts of sulfur trioxide are effective sulfonating agents.^[15]

Preparation

The direct oxidation of sulfur dioxide to sulfur trioxide in air proceeds very slowly:



Industrial

Industrially SO_3 is made by the contact process. Sulfur dioxide is produced by the burning of sulfur or iron pyrite (a sulfide ore of iron). After being purified by electrostatic precipitation, the SO_2 is then oxidised by atmospheric oxygen at between 400 and 600 °C over a catalyst. A typical catalyst consists of vanadium pentoxide

[illegible]

(monohydrate):

Interactive image

<https://chemapps.stolaf.edu/jmol/jmol.php?model=O>

O%3DS%28%3D
O%29%3DO)

<u>ChEBI</u>	CHEBI:29384 (https://www.ebi.ac.uk/chebi/searchId.do?chebiId=29384) ✓
<u>ChemSpider</u>	23080 (https://www.chemspider.com/Chemical-Structure.23080.html) ✓
<u>ECHA InfoCard</u>	100.028.361 (https://echa.europa.eu/substance-information/-/substanceinfo/100.028.361)
<u>EC Number</u>	231-197-3

[illegible]

Sulfur trioxide is a reagent in sulfonation reactions. These processes afford detergents, dyes, and pharmaceuticals. Sulfur trioxide is generated in situ from sulfuric acid or is used as a solution in the acid.

B₂O₃ stabilized sulfur trioxide was traded by Baker & Adamson under the tradename "*Sulfan*" in the 20th century.^[19]

Safety

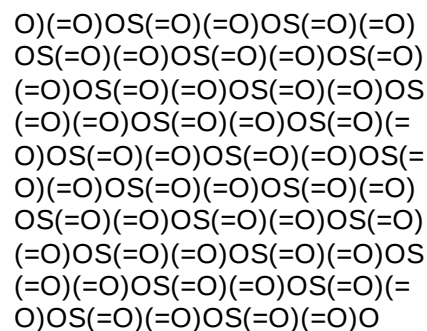
Along with being an oxidizing agent, sulfur trioxide is highly corrosive. It reacts violently with water to produce highly corrosive sulfuric acid.

See also

- Hypervalent molecule
- Sulfur trioxide pyridine complex

References

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(monohydrate): O.O=S(=O)=O

Properties	
Chemical formula	SO ₃
Molar mass	80.066 g/mol
Appearance	Colorless to white crystalline solid which will fume in air. ^[2] Colorless liquid and gas. ^[3]
Odor	Varies. Vapor is pungent; like sulfur dioxide. ^[4] Mist is odorless. ^[3]
Density	1.92 g/cm ³ , liquid
Melting point	16.9 °C (62.4 °F; 290.0 K)
Boiling point	45 °C (113 °F; 318 K)
Solubility in water	Reacts to give sulfuric acid
Thermochemistry	
Std molar entropy (S [⊖] ₂₉₈)	256.77 J·K ^{−1} ·mol ^{−1}
Std enthalpy of formation (Δ _f H [⊖] ₂₉₈)	−395.7 kJ/mol
Hazards	
Occupational safety and health (OHS/OSH):	
Main hazards	Highly corrosive, extremely strong dehydrating agent
GHS labelling:	

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Pictograms	
Signal word	Danger
Hazard statements	H314, H335
Precautionary statements	P261, P280, P305+P351+P338, P310 ^[5]
NFPA 704 (fire diamond)	
Flash point	Non-flammable
Lethal dose or concentration (LD, LC):	
LC ₅₀ (median concentration)	rat, 4 hr 375 mg/m ³
Safety data sheet (SDS)	ICSC 1202 (http://web.archive.org/web/20111019000350/http://incchem.org/documents/icsc/icsc/eics1202.htm)
Related compounds	
Other cations	Selenium trioxide Tellurium trioxide
Related sulfur oxides	Sulfur monoxide Sulfur dioxide
Related compounds	Sulfuric acid
Except where otherwise noted, data are given for materials in their standard state (at 25 °C [77 °F], 100 kPa). ✗ verify (what is ✓ ✗ ?) Infobox references	

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19. Habashi, Fathi; Dugdale, Raymond (June 1973) [1972-11-06]. "The Action of Sulfur Trioxide on Chalcopyrite" (<https://reader.zlibcdn.com/articles/6427f27508dc5960e6c75b3feac71828/1.jpg>). *Metallurgical and Materials Transactions*. **B-4** (6): 1553–1556. Bibcode:1973MT.....4.1553H (<https://ui.adsabs.harvard.edu/abs/1973MT.....4.1553H>). doi:10.1007/BF02668007 (<https://doi.org/10.1007%2F02668007>). S2CID 93744787 (<https://api.semanticscholar.org/CorpusID:93744787>). p. 1553: "Sulfur trioxide used was pure, colorless liquid SO₃ marketed under the trade name Sulfan by Baker and Adamson"

Sources

- NIST Standard Reference Database (<http://webbook.nist.gov/chemistry/>)
 - ChemSub Online (http://chemsub.online.fr/name/sulfur_trioxide.html)
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